

Consider the following linear problem:

$$\min_{\mathbf{x} \in \mathbb{R}^3} 2x_1 + 7x_2 + 3x_3$$

subject to

$$-2x_1 - x_2 + x_3 \geq 1 \quad (y_1)$$

$$x_1 + 2x_2 \geq 2 \quad (y_2)$$

$$x_1, x_2, x_3 \geq 0.$$

and its dual

$$\max_{\mathbf{y} \in \mathbb{R}^2} y_1 + 2y_2$$

subject to

$$-2y_1 + y_2 \leq 2 \quad (x_1),$$

$$-y_1 + 2y_2 \leq 7 \quad (x_2),$$

$$y_1 \leq 3 \quad (x_3),$$

$$y_1, y_2 \geq 0.$$

1. Solve the dual problem with the graphical method.
2. State the complementarity slackness conditions for both primal and dual problems. Using these, determine the optimal solution to the primal problem.
3. Verify the result that claims that the optimal solution to the dual problem satisfies

$$\mathbf{B}^T \mathbf{y}^* = \mathbf{c}_B,$$

where $\mathbf{B}$ is the basis matrix of the primal problem.